MATHEMATICS
Introduction to Modern Algebra I
Quiz 9 Solution.

1) List all possible abelian groups of order 200.
$200 = 2^3 5^2$. The possibilities are:
$\mathbb{Z}_8 \times \mathbb{Z}_{25}$,
$\mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_{25}$,
$\mathbb{Z}_2^3 \times \mathbb{Z}_{25}$,
$\mathbb{Z}_8 \times \mathbb{Z}_5^2$,
$\mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_5^2$ and
$\mathbb{Z}_2^3 \times \mathbb{Z}_5^2$.

2) $\mathbb{Z}_{12} \times \mathbb{Z}_2 \times \mathbb{Z}_{18}$ is isomorphic to which one of the following groups:
$\mathbb{Z}_2^4 \times \mathbb{Z}_3 \times \mathbb{Z}_9$, $\mathbb{Z}_{36} \times \mathbb{Z}_{12}$, $\mathbb{Z}_{36} \times \mathbb{Z}_6 \times \mathbb{Z}_2$, $\mathbb{Z}_{432}$, or $\mathbb{Z}_2^4 \times \mathbb{Z}_3^3$.
Answer: $\mathbb{Z}_{12} \times \mathbb{Z}_2 \times \mathbb{Z}_{18} \cong \mathbb{Z}_3 \times \mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_9 \cong \mathbb{Z}_{36} \times \mathbb{Z}_6 \times \mathbb{Z}_2$.